

Kaustav Mitra

Curriculum Vitae [abridged version]

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Education

- **Ph.D.**, *Yale University*, thesis advisor: [Frank C. van den Bosch](#) 2025
- **M.Sc. & M.Phil., Astrophysics**, *Yale University*, Ph.D. coursework: [\[link \]](#) 2021
- **B.Sc. & M.Sc., Physics**, *Presidency University* (Kolkata, India), coursework: [\[link \]](#) 2018

Selected Talks ($\diamond \Rightarrow$ invited talk)

- Variant of: “Differentiable forward-model of galaxy and halo assembly meets modern survey data”
 - $\diamond$ [MockNYC Conference](#), Flatiron Institute / Simons Foundation, New York (USA). Jan 2026
 - Early Career Scientist Symposium, [High Energy Phys., Argonne National Lab](#) (USA). Jan 2026
- Variant of: “Dynamical modelling of satellite galaxies to infer galaxy-halo connection and cosmology”
 - $\diamond$ Astronomy Seminar, [Department of Physics & Astronomy, Clemson University](#), (USA). Feb 2026
 - [AAS Meeting \(247th\)](#), Phoenix, AZ (USA). Jan 2026
 - $\diamond$ Galaxy Formation Talk, [Academia Sinica \(ASIAA\)](#), Taipei City (Taiwan). Feb 2025
 - $\diamond$ KIAA-PKU Seminar, [Kavli Institute for Astronomy and Astrophysics](#), Beijing (China). Dec 2024
 - [Santa Cruz Galaxy Workshop](#), University of California, Santa Cruz (USA). Aug 2024
 - [N3AS Summer School in Multi-Messenger Astrophysics](#), UCSC, California, (USA). Aug 2022
 - [53rd annual meeting, Division on Dynamical Astronomy \(DDA\)](#), CCA, NY (USA). April 2022
- Variant of: “Disk-Jet connection in blazars: modelling radiative transfer in multi-zone turbulent jets”
 - [Advances in Astroparticle Physics & Cosmology](#), SINP, WB (India). Jan 2020
 - The 29th New England Regional Quasar/AGN Meeting, MIT Kavli Institute, MA (USA). May 2019
- Variant of: “Halo Occupation of Quasars: AGN Unification From a Cosmological Perspective”
 - [APS April meeting \(Distinguished Student Fellowship\)](#), Columbus, OH (USA). April 2018
 - [36th meeting, Astronomical Society of India](#), Hyderabad (India). Feb 2018

Teaching Experience

At Yale University (see the long-version CV for past teaching experiences)

- **Teaching Assistantship in undergraduate courses**
 - Galaxies and the Universe (ASTR 120) *main instructor*: Robert Zinn
 - Introduction to Observational Astronomy (ASTR 155) *main instructor*: Michael Faison
 - Planets and Stars (ASTR 110) *main instructor*: Robert Zinn
 - Galaxies and the Universe (ASTR 120) *main instructor*: Jeffrey Kenney
 - Origins and the Search for Life in the Universe (ASTR 130) *main instructor*: Debra Fischer

Scholarships and Awards

1. 3 consecutive times awardee of Yale Dean’s Colloquium fund, as the organiser of Yale cosmology seminar series. 2022, 2023, 2024
2. [2nd prize winner of Yale 3-minute thesis \(3MT\) competition in the physical sciences division](#) 2023
3. [American Physical Society Distinguished Student Fellowship](#) 2018

4. **S. N. Bhatt Memorial Excellence Fellowship**, for undergraduate summer research, June - Aug, 2017
at the International Centre for Theoretical Sciences (ICTS-TIFR), India.
5. **Kishore Vaigyanik Protsahan Yojana (KVPY) scholarship**, 2013 - 2018
Department of Science and Technology, Government of India.
6. **DST-INSPIRE Fellowship**, Department of Science and Technology, Government of India. 2013
7. **Jagadis Bose National Science Talent Search (JBNSTS) scholarship**. 2013

Outreach

Public Talks

- **Science In The News** (Talks in public libraries, partnering with lifelong-learners' programs) 2024
"Ray tracing from computers to the cosmos" (jointly with 2 computer science PhD students)
 – Essex library, 33 West Avenue, Essex, CT 06426 May 11th
 – Mark Twain library, 439 Redding Road, Redding, CT 06896 May 21st
 – Norwich public library, 368 Main St. Norwich, VT 05055 May 30th
 – **Schiller Shoreline Institute for Lifelong Learning**, May 28th
 Nathanael B. Greene Community Center 32 Church St, Guilford, CT 06437
- **"Using high-school physics to unlock the strangest mysteries of the universe"** (some variation of this)
 – 2 pedagogical classes for high school students from Connecticut (**Yale Splash initiative**). Nov 2023
 – 1 hour lecture, addressing high school students of St. Timothy's School, Maryland. March 2021
 – **Yale Pathways to Science** for high school students from Connecticut. (20 minute talk) June 2020
 – **Open Labs Exploring Science week** for middle and high school students of greater New Haven area. Feb 2020
- "What are stars made of?", **Open Labs Science Cafe** talk for middle and high school students. Oct 2020
- "How to weigh galaxies and why do we need to?", **Astronomy on Tap, New Haven**, CT (USA). Feb 2020

Leitner Family Observatory and Planetarium

- I have regularly presented planetarium shows at Yale's Leitner Family Observatory and Planetarium, and have led numerous stargazing sessions there during the public-nights, every Tuesdays. 2019, 2020

Professional Commitments and Leadership Roles

- Mentor on **Project SHORT**: a student-led non-profit organisation guiding students from underprivileged backgrounds, mostly from developing countries, on graduate school applications. 2022 - Now
- I started the weekly Yale Cosmology Seminars and I was the main organizer for 2 years. 2022 - 2024
- PhD student interviewer in graduate student recruitment. 2022 - 2024
- Content Editor, Yale Department of Astronomy Newsletter. 2023
- Co-organizer of weekly Galaxy Lunch seminars, Yale Department of Astronomy. 2022 - 2023
- Graduate student representative in the prospective student webinar and information session. Nov 2022
- Served in the student panel for interviewing and evaluating shortlisted applicants in the departmental faculty hiring process. 2022
- Two times "Data tsar" at CHIME telescope (one responsible for monitoring the smooth running of the telescope) 2019 - 2020
- Student member of the American Physical Society and member of American Physical Society (APS) Forums On Graduate Student Affairs (FGSA) and Outreach and Engaging the Public (FOEP). 2017 - 2019
- Science communication and public outreach volunteer
American Physical Society (APS) April meeting, Columbus, Ohio, USA. April 2018
- Member of scientific organising committee and session chair of astrophysics session **Undergraduate physics symposium**, Presidency University. April 2017
- Student Coordinator, **'Advanced School on Gravitational Waves'**, Presidency University December 2016

List of Publications

Refereed Papers

1. [Kaustav Mitra](#), Frank C. van den Bosch, Josephine Baggen, Johannes U. Lange, (in rev.) “[BASILISK IV. No \$S_8\$ Tension with Satellite Kinematics](#)”. OJAp [in review]
2. [Kaustav Mitra](#), Frank C. van den Bosch, (in rev.) “[BASILISK III. Stress-testing the Conditional Luminosity Function model](#)”. OJAp [in review]
3. Josephine Baggen, Frank C. van den Bosch, [Kaustav Mitra](#), (in rev.) “[The Impact of Baryonic Effects on the Dynamical Masses Inferred Using Satellite Kinematics](#)”. OJAp [in review]
4. J. Sebastian Monzon, Frank C. van den Bosch, [Kaustav Mitra](#) (2024). “[Constraining the low mass end of the stellar halo mass relation with surveys of satellite galaxies](#)”. ApJ, Volume 976, Issue 2, id.197, 23 pp.
5. [Kaustav Mitra](#), Frank C. van den Bosch, Johannes U. Lange, (2024) “[BASILISK II. Improved constraints on the galaxy-halo connection from satellite kinematics in SDSS](#)”. MNRAS, 533(3), 3647-3675
6. Aritra Ghosh, et al., including [Kaustav Mitra](#) (2024). “[Denser environments cultivate larger galaxies: a comprehensive study beyond the Local Universe with 3 million Hyper Suprime-Cam galaxies](#)” 971(2), id.142, 23 pp.
7. Pieter van Dokkum, et al., including [Kaustav Mitra](#) (2023). “[A Candidate Runaway Supermassive Black Hole Identified by Shocks and Star Formation in its Wake](#)”. ApJL, 946(2), 14 pp.
8. Saugata Barat, Ritaban Chatterjee, [Kaustav Mitra](#) (2022). “[Locating the GeV emission region in the jets of blazars from months time-scale multiwavelength outbursts](#)”. MNRAS, 515(2), 1655-1662.
9. Aritra Kundu, Ritaban Chatterjee, [Kaustav Mitra](#), Sripan Mondal (2022). “[RMS-flux relation and disc-jet connection in blazars in the context of the internal shocks model](#)”. MNRAS, 510(3), 3688-3700.
10. Anwesh Majumder, [Kaustav Mitra](#), Ritaban Chatterjee, C. Megan Urry, Charles Bailyn, Prantik Nandi (2019). “[Physical inference from the \$\gamma\$ -ray, X-ray, and optical time variability of a large sample of Fermi blazars](#)”. MNRAS, 490(1), 124-134.
11. Sagnick Mukherjee, [Kaustav Mitra](#), Ritaban Chatterjee (2019). “[The accretion disc-jet connection in blazars](#)”. MNRAS, 486(2), 1672-1680.
12. [Kaustav Mitra](#), Suchetana Chatterjee, Michael A. DiPompeo, Adam D. Myers, Zheng Zheng (2018). “[The Halo Occupation Distribution of obscured quasars: revisiting the unification model](#)”. MNRAS, 477(1), 45-55.

Conference Proceedings

1. [Kaustav Mitra](#), Frank C. van den Bosch, Josephine Baggen, Johannes U. Lange (2025). “[Novel probe of \$S_8\$ tension: insensitive to halo assembly bias and incorporating baryonic effects](#)”. 245th Meeting of the American Astronomical Society, id. 250.25. Bulletin of the American Astronomical Society, Vol. 57, No. 2.
2. [Kaustav Mitra](#), Frank C. van den Bosch (2022). “[Dynamical modelling of satellite galaxies to infer galaxy-halo connection and cosmology](#)”. AAS Division on Dynamical Astronomy meeting #53, id. 405.04. Bulletin of the American Astronomical Society, Vol. 54, No. 4.
3. Adam D. Myers, Michael A. DiPompeo, [Kaustav Mitra](#), Ryan C. Hickox, Suchetana Chatterjee, Kelly Whalen (2018). “[Characterizing the evolution of WISE-selected obscured and unobscured quasars using HOD models](#)”. American Astronomical Society, AAS Meeting #232, id. 322.01.
4. [Kaustav Mitra](#), Suchetana Chatterjee, Michael A. DiPompeo, Adam D. Myers, Zheng Zheng (2018). “[Halo Occupation of Quasars: AGN Unification From a Cosmological Perspective](#)”. American Physical Society (APS) April Meeting 2018, abstract id.S16.001